FINLITE: A WEB-BASED FINANCIAL TRACKER AND AUTOMATIC REPORT GENERATION SYSTEM WITH AI CHATBOT FOR THE LEAGUE OF INFORMATION TECHNOLOGY ENTHUSIASTS

A Project
Presented to the Faculty of the
College of Computer Studies
Pambayang Dalubhasaan ng Marilao

In Partial Fulfillment
of the requirements for the
System Analysis and Design

By

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1. INTRODUCTION

This chapter presents the first part of the study, introducing the project context by outlining the proposed system, the target organization's background, the business logic and operational processes of the current system, and the specific problems encountered in existing manual operations.

1.1. Project Context

In contemporary organizational management, financial accountability hinges on the systematic recording of monetary flows and the punctual delivery of audit-compliant liquidation reports. A financial tracker serves as a centralized digital mechanism that systematically logs, categorizes, and monitors financial inflows and disbursements in real time, establishing operational transparency, strict custodial accountability, and verifiable audit trails. Complementing this is an automatic report generation system, an automated data processing engine that programmatically compiles, cross-verifies, and formats transactional data into standardized, audit-ready institutional reports, completely removing the vulnerabilities of manual calculation, arithmetic errors, and formatting non-compliance. FinLITE: A Web-Based Financial Tracker and Automatic Report Generation System with AI. Chatbot integrates these two core functionalities into an operational financial co-pilot tailored to the unique administrative dynamics of the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng Marilao (PDM).

Pambayang Dalubhasaan ng Marilao (PDM) is a local tertiary institution established pursuant to municipal ordinances to provide accessible, high-quality higher education to the youth of Marilao and surrounding Bulacan municipalities, operating in compliance with the regulatory standards of the Commission on Higher Education (CHED) and the Association of Local Colleges and Universities (ALCU). Within PDM, the College of Computer Studies (CCS) is the primary computing division, offering the flagship Bachelor of Science in Information Technology (BSIT) program under the executive academic leadership of Dean Dr. Emraida Marie M. Manucom and BSIT Program Director Jovylyn Ortiz-Cesar, MBA, MSIT.

The League of Information Technology Enthusiasts (LITE) is the officially accredited academic mother organization representing all students enrolled in the BSIT program at PDM. Guided by Club Advisers Ms. Kimberly Dawn Jatulan and Ms. Krizia Mae Genovia, LITE organizes college-wide technical seminars, programming hackathons, e-sports competitions, social development initiatives, and official participation in campus events such as Institutional Club Week. LITE is governed by a multi-tiered leadership structure comprising an Executive Board—composed of the President, Vice President, Secretary, Treasurer, and Auditor—elected Year-Level

Representatives for the 1st, 2nd, 3rd, and 4th Year cohorts, and nine specialized sub-committees covering Arduino, Cybersecurity, Programming, Networking, Multimedia, Photography, E-sports, Singing, and Dancing disciplines.

LITE operates under severe collegiate physical constraints, possessing no dedicated physical office, private conference room, or institutional workstation. Administrative planning, audit sessions, and ledger consultations take place transiently in campus corridors, student lounges, event booths, or at the adviser's desk located in the second-floor faculty room. Money custody follows strict institutional protocols where all physical cash holdings, petty cash reserves, and accumulated coin collections are secured inside a single small metal cashbox stored inside a locked drawer within the Program Director's Room. The small metal cashbox never leaves the PDM campus. During off-campus purchasing expeditions for event supplies, officers usually get the budget from the cash box with authorization from club advisers. Purchases are only funded out of pocket via personal advances of officers or advisers if there is not enough budget inside the cash box, which are reimbursed upon return to campus. Furthermore, leadership transitions occur annually under a strict face-value turnover protocol, where the physical cash in the small metal cash box must match the exact numerical balance attested on the audited final financial report before outgoing officers receive institutional clearance. Legitimate unpaid reimbursement claims remaining at the end of the academic year are certified and carried over as Accounts Payable liabilities to the incoming administration.

Under certified institutional policy, LITE does not collect mandatory student membership dues. Operating capital originates from four legitimate revenue streams: student entry fees from competitive departmental events and tournaments; batch merchandise orders such as departmental lanyards and organization T-shirts, which strictly enforce one-hundred percent upfront payment prior to vendor ordering; booth rental fees and sales commissions collected from concessionaires during Institutional Club Week; and audited carried-over surplus reserves alongside certified opening liabilities from preceding academic years.

Collection of funds traverses two distinct operational channels. Physical cash collected at event booths by the Treasurer, Year-Level Representatives, or Sub-Committee Chairpersons is subject to a strict holding window, expected by the end of the day or the day after the collection to be turned over to the Treasurer for deposit into the small metal cashbox in the Program Director's room, though in practice this window is frequently delayed during intensive multi-day events. During active event booths, representatives may execute emergency supply deductions directly from unremitted collections, provided an official receipt or signed petty cash slip is submitted immediately upon turnover. Digital payments for tournament entries and merchandise orders are

captured through the personal GCash mobile wallet of the Treasurer or on-duty executive officers such as the President. Third-party withdrawal service fees based on the store's service charge are logged as an operational expense labeled Bank or Cash-Out Fee, while odd fractional coin balances that cannot be withdrawn physically remain settled into the officer's personal account.

Disbursements for event supplies, certificates, sashes, trophies, and refreshments are governed by spending authorization thresholds. Any expenditure exceeding one thousand pesos strictly requires prior written or chat approval from both Club Advisers, whereas purchases below this threshold proceed at executive officer discretion. Off-campus purchasing runs rely on out-of-pocket personal capital advances, termed Abono, by club advisers and student officers. These advances represent unliquidated organizational liabilities under Accounts Payable, requiring tracking of both the original payer and the designated claimant requesting reimbursement.

Liquidation of personal advances requires receipt submission within a strict seven-day grace period following activity completion, although strict enforcement on club advisers is often informal. Commercial purchases require official printed commercial receipts or sales invoices. Non-receipted micro-disbursements, such as tricycle fares, public market craft purchases, or emergency printouts, require a handwritten petty cash acknowledgment slip detailing the date, monetary amount, travel route or item purpose, and claimant signature. If approved reimbursement claims exceed available physical cashbox liquidity, a partial cash payout is executed, disbursing available cash while logging the residual balance as an active pending liability.

Physical cash reconciliations are conducted periodically at the adviser's desk in the second-floor faculty room by sorting banknotes and coins from one thousand pesos down to one-peso coins with official witnesses present. The theoretical ledger balance is calculated by adding opening carryover balances and inflows, then subtracting reimbursed disbursements. The discrepancy variance is calculated by subtracting the theoretical balance from total counted physical cash. Balanced counts are certified at zero variance. If a shortage occurs, all organizational spending and club activities immediately freeze until investigated. Untraceable minor shortfalls may be declared as an allowable operating expense labeled Cash Shortage under written justification and dual adviser sign-off, as exemplified by a historical one-hundred sixty-one peso shortage declaration. Cash overages are retained inside the cashbox as unrecorded emergency petty cash buffers.

At semester end, LITE prepares a Final Financial Liquidation Report formatted in Times New Roman typography with borderless tables. Following digital draft review by Club Advisers, hard copies printed on institutional bond paper are routed sequentially

across a six-stage wet-ink signatory chain: starting with the Treasurer; followed by the Auditor; moving to the President; proceeding to both Club Advisers for joint dual sign-off; moving to the BSIT Program Director for departmental review; and concluding with the Dean of the College of Computer Studies for final executive approval. If formatting or calculation errors are identified at later stages, the report is rejected, requiring manual re-editing, re-printing, and restarting the physical six-stage routing chain from the beginning.

Primary field interviews with executive officers across successive administrations from 2024 to 2026, corroborated by adviser questionnaires, reveal six critical operational problems within LITE's current manual financial workflows.

The first problem is personal financial vulnerability and delayed reimbursement of personal advances. Because LITE cannot maintain pre-disbursed operational bank accounts, club advisers and student executive officers routinely advance thousands of pesos of personal funds for competition tokens, food, certificates, and booth materials. Thermal paper receipts gathered during fast-paced events frequently fade, crumple, or get lost. Furthermore, late submission of receipts by student officers past the seven-day deadline leaves reimbursement queues stalled for weeks, clouding organizational liabilities. Loose receipt slips also lack metadata clarifying who advanced the funds versus who is claiming reimbursement. When receipts are lost or submitted past deadlines without prior approval, the expenditures cannot be audited, forcing advisers and officers to absorb these expenses out-of-pocket.

The second problem is the commingling of personal and organizational capital via ad hoc GCash pooling. Relying on personal mobile wallets of the Treasurer or on-duty officers to capture digital collections for tournament fees, merchandise orders, and booth rentals creates severe internal control risks. Organizational funds co-mingle with personal savings, raising accidental spending risks. Temporal lag between digital collection and physical cash-out produces ledger drift, where digital records reflect collected funds but the physical cashbox lacks corresponding cash. Unlogged withdrawal fees of fifteen to twenty pesos create compounding cash deficits, while odd fractional balances remain trapped in personal digital wallets due to ATM denomination limits.

The third problem is ephemeral documentation and administrative leakage in non-receipted micro-disbursements. Non-receipted micro-disbursements, such as jeepney and tricycle fares, local market craft purchases, and emergency photocopies, are currently documented through loose paper scraps, sticky notes, or unstructured Messenger chat screenshots. These informal records lack standardized metadata, including exact timestamps, itemized descriptions, authorized claimant identifiers, and supervising officer endorsements. Over a multi-month semester, these paper scraps are smudged or mislaid,

depriving the Auditor of verifiable audit trails and resulting in unliquidated expense leakage.

The fourth problem encompasses physical cashbox custodial risks and high-stress face-value turnover obligations. Storing physical funds in a small metal cashbox inside a locked drawer in the Program Director's room presents custodial exposure as cash is transported across corridors during active campus events. Furthermore, holding collection funds for twenty-four to forty-eight hours creates temporary visibility blind spots. Under the mandatory strict face-value turnover protocol, incoming administrations refuse custody if physical cash deviates by even a single peso from the final audited balance. Because transaction tracking is conducted post-hoc on loose paper, historical errors accumulate invisibly, compelling outgoing officers to pay out-of-pocket to rectify balance discrepancies before receiving graduation or leadership clearance.

The fifth problem involves latent discrepancies and arithmetical reconciliation bottlenecks. Physical cash counts, sorting bills from one thousand pesos down to one-peso bills, are performed weeks or months after events conclude rather than on a daily cadence. In the absence of automated reconciliation tools, officers perform manual cross-multiplication against loose paper tallies, hiding arithmetic errors until final semester report compilation. When cash variances emerge, the resulting mandatory spending and activity freeze paralyzes club operations while officers struggle to determine whether the disparity represents an unrecorded GCash cash-out, an unlogged petty disbursement, or an untraceable coin variance.

The sixth problem consists of institutional liquidation bottlenecks and signatory rejections. Manual preparation of end-of-term liquidation reports using basic word processors results in formatting non-compliance with College of Computer Studies standards, including uneven table borders, misaligned decimal figures, and inconsistent font sizes. Navigating the physical six-stage wet-ink signatory chain, including the mandatory dual adviser joint sign-off, becomes an acute operational bottleneck. If a calculation error, missing voucher attachment, or typographical defect is caught by the Program Director or Dean at the fifth or sixth stage, the report is rejected. Under manual operations, officers must manually re-compute, re-edit, reprint the multi-page document on institutional bond paper, and restart physical wet-ink routing from the first stage, causing immense administrative gridlock during final examination and graduation periods.

1.2. Objectives of the Study

1.2.1. General Objectives

The general objective of the study is to design and develop FinLITE: A Web-Based Financial Tracker and Automatic Report Generation System with an AI Chatbot for the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng Marilao to streamline organizational financial transaction tracking, automate institutional liquidation report compilation, and provide intelligent audit assistance.

1.2.2. Specific Objectives

This project has the following specific objectives:

1. 1. To design a secure web-based financial tracker that records cash collections, digital GCash payments, and personal advances (*abono*) to prevent lost funds and mixed accounts.

2. To create a digital cash denomination counter and automated reconciliation tool that tallies physical bills and coins against ledger balances to eliminate math errors and prevent operational freezes.

3. To provide an automated report generation engine that creates CCS-compliant Word liquidation reports and an AI Chatbot that answers budget and expense questions instantly.

1.3. Purpose and Description

This study is significant to the following:

1.3.1 LITE Organization Members and Officials:

The system benefits student leaders and members by replacing paper records with a digital ledger, ensuring fast reimbursement of personal advances (*abono*), and protecting outgoing officers from paying cash shortages out-of-pocket during turnover.

1.3.2 Club Advisers:

The system benefits advisers by giving them real-time visibility over club funds, making it easy to approve expenses over ₱1,000, and allowing them to ask the AI Chatbot for instant budget updates.

1.3.3 BSIT students:

The system will benefit the general student body of the Bachelor of Science in Information Technology program by upholding financial integrity and accountability in the management of student activity fees, merchandise payments, and event booth

revenues, thereby ensuring that organizational funds are utilized efficiently for meaningful collegiate technical activities and student development.

1.3.4 BSIT Program Administration and College Leadership:

The system benefits the Program Director and the Dean by ensuring all submitted financial reports follow correct CCS formatting and math, avoiding repeated document rejections and delays.

1.3.5 The Proponents:

The system benefits the researchers by helping them apply their skills in web programming, database design, and AI integration to solve a real campus problem.

1.3.6 Future Researchers:

The system benefits future researchers by serving as a useful guide and reference for building financial tools for student organizations.

1.4. Scope and Delimitation

The scope of this study covers the design, development, and implementation of FinLITE: A Web-Based Financial Tracker and Automatic Report Generation System with AI Chatbot for the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng Marilao (PDM). The proposed system is structured around an integrated user journey that begins at account authentication and systematically guides users through dedicated operational modules, with each module purposefully arranged to resolve the specific financial problems identified in the organization's existing manual workflows.

The system workflow initiates with the User Authentication and Role-Based Access Control Module, where the LITE president and vice president, treasurer, auditor, and faculty club advisers securely register and log in to the web platform. Upon credential verification, the club advisers become the system administrators and assign distinct operational permissions according to designated organizational roles. The Treasurer gains permissions to encode cash collections and upfront merchandise payments into the ledger, log GCash withdrawals along with their convenience fees, record advance reimbursement payouts, operate the digital denomination keypad during physical cash counts, and generate one-click CCS-compliant liquidation reports. The Auditor receives read-only verification access to inspect uploaded receipt photos and vouchers without altering records, monitor missing documentation in real time, review automated variance flags, record approved shortage justifications or cashbox buffers, and execute digital audit sign-offs. Meanwhile, the LITE President and Vice President obtain

rights to track real-time organizational budget balances, liquidate personal mobile wallet collections, preview and edit report text directly in-browser, execute executive digital sign-offs, and unlock returned reports for quick re-export. The Faculty Club Advisers maintain administrative oversight, accessing real-time dashboard monitoring of liquid assets and pending debts, digital approval controls for expenditures exceeding the one thousand peso threshold, joint dual sign-off capabilities for finalized reports, and natural-language inquiry access to the AI Chatbot co-pilot. Additionally, authorized executive officers hold permissions to log event booth collection remittances under custodial tracking and generate electronic petty cash vouchers with preset travel routes for official campus errands.

Once logged in, the Club Adviser accesses the Advance & Reimbursement Management Module to resolve personal financial vulnerability and delayed reimbursement of out-of-pocket advances (abono). When an officer or Club Adviser spends personal money due to a lack of organizational funds for competition tokens, certificates, food, or booth materials, they open this module on their smartphone or computer to file an immediate claim. Using a dual-entity tracking form, the claimant designates both the original payer and the designated claimant, then takes a photo and uploads the commercial sales invoice. The module automatically initiates a seven-day liquidation countdown timer with visual status badges (Active, Warning, Overdue) before thermal receipt ink fades or slips get lost. Through a restricted interface, the Auditor performs real-time receipt tracking and read-only verification to inspect the uploaded proofs and catch missing claimant details early, while the Club Adviser executes digital approval of the abono claim. Meanwhile, the LITE President or Vice President monitors the outstanding reimbursement debt on the budget dashboard, and the Treasurer accesses the abono payout queue to execute cashbox reimbursements, systematically logging full or partial payout states and updating creditor balances to ensure no leader absorbs organizational expenses out of pocket.

Following personal advance tracking, the Treasurer and LITE President or Vice President navigate to the Dual-Vault Balance Tracker and Cash-Out Logging Wizard to resolve the mix of personal and organizational funds caused by informal GCash payment collection. For digital collections gathered during tournament registrations or upfront merchandise orders, the Treasurer encodes the transactions into the ledger while the LITE President or Vice President executes GCash liquidation by recording mobile wallet payment proofs directly into the platform, safely isolating the funds inside a Digital GCash In-Transit Vault. When the Treasurer or LITE President/Vice President withdraws the collected funds at an over-the-counter merchant or automated teller machine, they run the Cash-Out Logging Wizard. The withdrawing officer uploads the transaction screenshot, records the exact physical cash retrieved, and logs the ₱15.00 to ₱20.00 withdrawal fee as a legitimate operational expense labeled Bank/Cash-Out Fee. Any odd,

unwithdrawable centavos are automatically transferred to a digital fractional coin buffer, maintaining exact mathematical parity between digital records and physical cash on hand, while the Club Advisers monitor the live cashbox and GCash vault balances in real time on their dashboard.

To address the third problem of temporary documentation and administrative leakage in non-receipted micro-disbursements, the purchasing officer—such as the LITE President/Vice President or Treasurer—accesses the Electronic Petty Cash Voucher (e-PCV) Generator. For low-value transit and incidental expenses that lack official commercial receipts, such as tricycle and jeepney fares between PDM and Marilao Poblacion, emergency photocopies, or public market craft purchases, the purchasing officer generates a digital voucher directly on screen. The officer selects from standardized travel route presets or expense categories, receives an auto-indexed voucher tracking number, inputs the itemized description, and captures their on-screen digital claimant signature alongside the supervising digital endorsement of the LITE President. Once submitted, the Auditor accesses the restricted interface to perform read-only verification, inspecting the voucher details and tracking claimant metadata in real time to prevent expense leakage, while the Treasurer records the corresponding cashbox disbursement, permanently eliminating fragile paper scraps, sticky notes, and unstructured chat screenshots.

The Treasurer and LITE President/Vice President then transition to the Witnessed Custodial Tracking and Turnover Module to resolve physical cashbox custodial risks and high-stress face-value turnover obligations. When event collections conclude, the system activates an automated holding-window timer that monitors physical funds held by on-duty collecting officers, dispatching digital alerts to the Treasurer, LITE President or Vice President, and Club Advisers if collections remain unremitted past the required 24-to-48-hour window before being locked in the metal cashbox in the Program Director’s Room. Upon remittance, the Treasurer encodes the turned-over cash into the single-entry ledger. At the conclusion of an academic tenure, the outgoing Treasurer and LITE President and Vice President run the turnover protocol witnessed by the Auditor and Club Advisers, generating an official Turnover Reconciliation Certificate that validates counted physical cash against audited ledger balances down to the exact peso. The module formally archives legitimate unsettled reimbursement claims as certified Accounts Payable carryover liabilities for the incoming administration, safeguarding outgoing leaders from unjustified out-of-pocket balance deductions during graduation clearance.

To eliminate latent arithmetic errors and prevent paralyzing club-wide spending freezes, the Treasurer and Auditor access the Interactive Cash Denomination Counter and Variance Reconciliation Engine during periodic physical audits conducted at the adviser's

desk. Instead of computing loose paper tallies by hand, the Treasurer inputs the physical quantity of each bill (₱1,000, ₱500, ₱200, ₱100, ₱50, ₱20) and coin denomination (₱20, ₱10, ₱5, ₱1) into the digital keypad calculator. The engine instantly computes the total cash on hand and cross-references it in real time against the theoretical ledger balance, triggering an automated variance alert for the Auditor if a discrepancy is detected. Under the discrepancy logging protocol, if a verified untraceable minor shortfall occurs, the system automatically drafts a standardized Cash Shortage Justification Report for the Auditor to log and the Club Advisers to endorse through digital dual sign-off. Conversely, any cash overage is credited directly to the emergency petty cash buffer, maintaining transparent accountability while preventing unwarranted operational freezes.

Finally, the Treasurer and LITE President/Vice President navigate to the one-click Institutional Document Generation Engine and AI Chatbot Co-Pilot to solve institutional liquidation bottlenecks, formatting non-compliance, and signatory rejections. With a single click, the Treasurer triggers the export engine, which pulls all validated ledger entries, electronic petty cash vouchers (e-PCVs) for non-receipted micro-expenses, and reconciliation logs to programmatically compile end-of-term financial reports in Microsoft Word (.docx) and PDF formats. These reports strictly comply with College of Computer Studies (CCS) standards, featuring Times New Roman typography, 12pt font sizing, borderless tables, right-aligned two-decimal currency figures, and the pre-formatted six-stage wet-ink signatory block. Directly in-browser, the LITE President or Vice President previews the exact report layout to make live text and metadata adjustments before export, and possesses the capability to unlock returned reports for quick re-export without re-typing the entire document from scratch. The report then progresses through the system's digital routing workflow, securing online certification from the Auditor at Stage 2, executive approval from the LITE President/Vice President at Stage 3, and digital dual sign-off from both Club Advisers at Stage 4 prior to physical printing for the Program Director and Dean. Concurrently, an embedded zero-hallucination AI Chatbot Co-Pilot is accessible across the platform, allowing executive officers and Club Advisers to submit natural-language inquiries to check remaining event budgets, monitor reimbursement debts, and inspect category totals instantly without manual spreadsheet combing.

The operational boundaries of this study are defined by several delimitations to preserve institutional compliance and feasibility. First, FinLITE does not integrate direct financial gateway APIs or banking webhooks for automated mobile wallet disbursements or fund transfers; digital payments and cash-out charges are recorded and verified manually by authorized officers through uploaded transaction proofs and reference numbers. Second, the system does not encompass automated physical currency handling, mechanical coin sorting, or electronic locking mechanisms for the physical metal cashbox

stored inside the Program Director's room, operating strictly as a digital ledger, calculation, and audit tracker.

Third, while the platform automates report calculation, layout formatting, and pre-audit validation, it does not bypass or replace the physical wet-ink signatory routing chain required by collegiate governance; final generated reports must still be printed on institutional bond paper and manually signed across all six administrative stages for official institutional filing. Lastly, the system's workflow rules, spending thresholds (such as the mandatory dual-adviser sign-off for expenses above ₱1,000.00), and reporting templates are customized exclusively for the League of Information Technology Enthusiasts (LITE) under the College of Computer Studies at Pambayang Dalubhasaan ng Marilao, and do not apply to other academic organizations, campus student councils, or external academic departments.

2. TECHNICAL BACKGROUND OF THE STUDY

2.1 Project Analysis

The analysis phase thoroughly investigated the existing financial tracking, cash custody, and liquidation routines of LITE across successive administrations from 2024 to 2026. Data gathered through questionnaires via Google Forms with the 2026-2027 LITE president and faculty advisers revealed critical inefficiencies in receipt handling, personal mobile wallet commingling, arithmetic reconciliation, and multi-stage hardcopy routing. To establish a rigorous architectural foundation, the operational workflows of both the existing manual system and the proposed FinLITE platform were formally modeled using structured Data Flow Diagrams (DFDs) following standard Gane-Sarson notation.

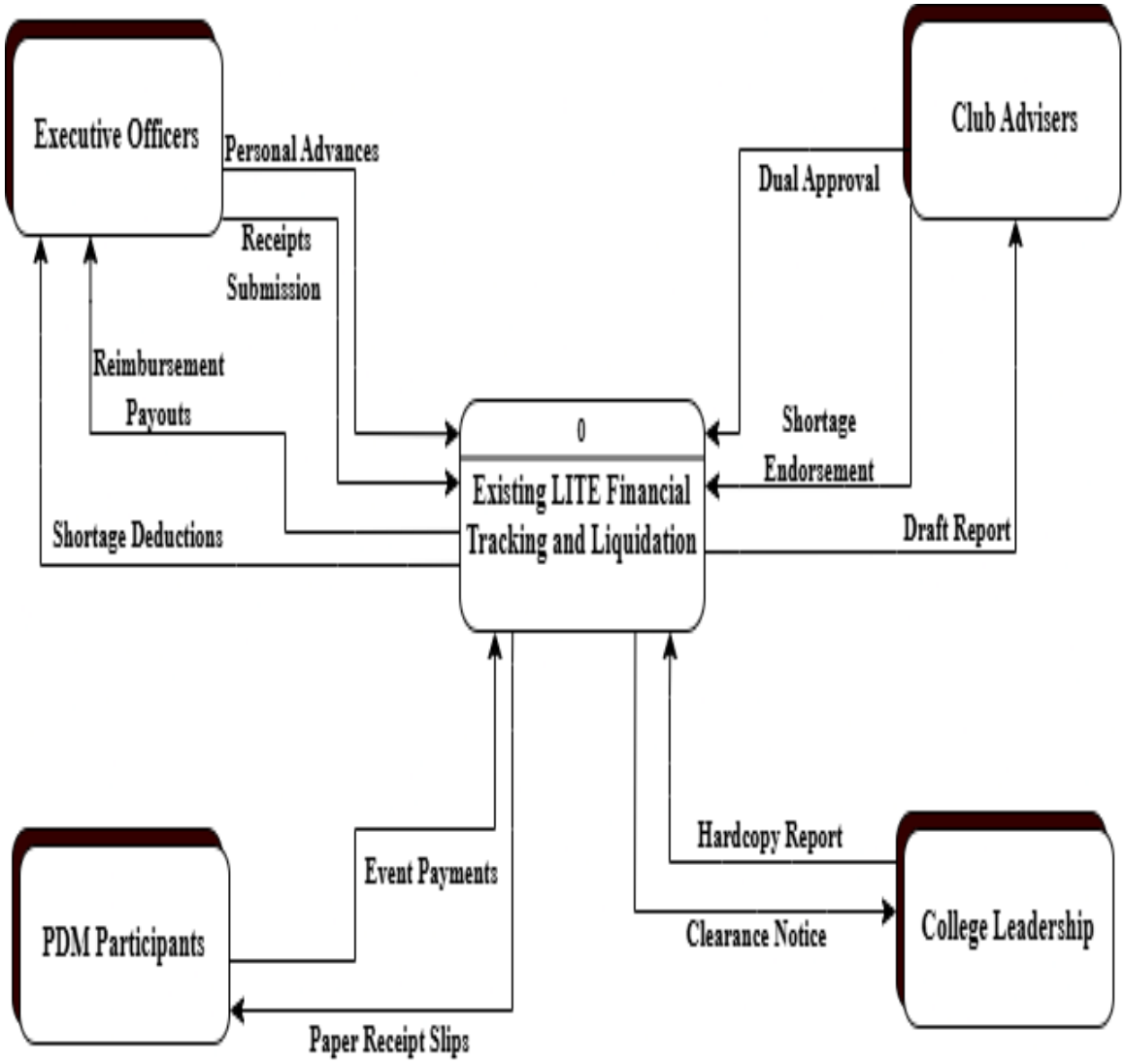


Figure 1: Context Diagram of the Existing LITE Financial Tracking and Liquidation

Figure 1 illustrates the Context Diagram of LITE Financial Tracking and Liquidation illustrates the high-level boundary and external information exchanges of LITE's existing manual financial workflow. In the current system, financial operations are driven by physical cash exchanges, loose paper documentation, and manual verbal or chat requests. When officers or faculty advisers execute emergency purchasing runs for competition materials, trophies, and food, they front personal capital (Abono) and gather thermal paper receipts, which are manually handed over to the Treasurer. Students and tournament participants hand over physical cash entry fees to on-duty officers, receiving handwritten paper acknowledgment slips or event wristbands in return.

Physical cash holdings are held by student collectors during events and later deposited into a single small metal cashbox secured inside a locked drawer in the BSIT Program Director's Room. Expenditures exceeding one thousand pesos require written or chat endorsement from both Faculty Club Advisers. At the conclusion of the academic semester, the Treasurer manually formats a multi-page liquidation report using desktop word processors, which must navigate a physical, sequential six-stage wet-ink signatory chain (Treasurer, Auditor, President, Dual Faculty Advisers, Program Director, and CCS Dean). Any calculation or typographical error detected by college administrators results in complete rejection, forcing manual re-editing, re-printing on bond paper, and restarting the entire six-stage routing chain from the beginning

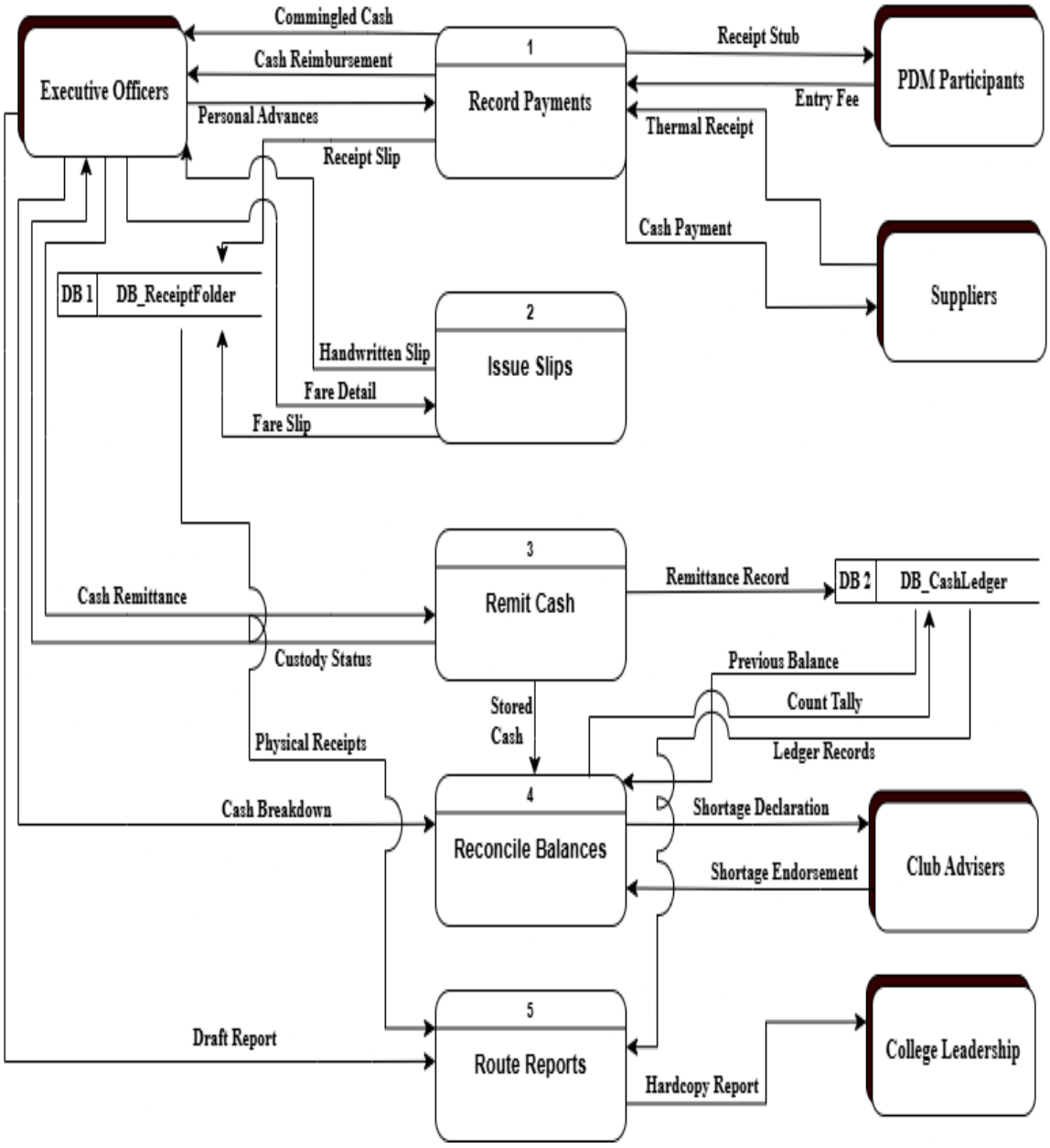


Figure 2: Existing Diagram 0 of LITE Financial Tracking and Liquidation

Figure 2 shows that whenever the student organization prepares for an event, executive officers usually use their own personal money first to buy supplies from local stores and vendors. In return, the suppliers hand them printed thermal receipts, which the officers keep inside a paper folder. At the same time, students participating in the event pay their registration fees in cash and receive a simple tear-off paper stub as proof of payment. Because officers collect this cash on the go, the club money often mixes with their personal funds in their pockets and wallets, which they later use to pay themselves back for what they spent in advance. For travel expenses like jeepney and tricycle rides where drivers cannot issue official receipts, the officers write down their fare details on handwritten slips and store them in the folder alongside the other receipts.

Once collections and errands are done, the officers hand over the collected cash to the treasurer. The treasurer notes down who is holding the funds, writes the remittance into a physical ledger notebook, and places the physical bills and coins into storage. When it is time to check the funds, the treasurer pulls out the stored cash and manually counts every bill and coin. This physical count is compared with the previous balance recorded in the ledger to see if the amounts match. If any money is missing, the officers have to write an explanation letter and submit it to their faculty advisers to get a signed endorsement. Finally, the treasurer prepares a draft report, gathers all the paper receipts from the folder, matches them with the records in the ledger, and prints out a final hardcopy report to submit to college leadership for institutional review and clearance

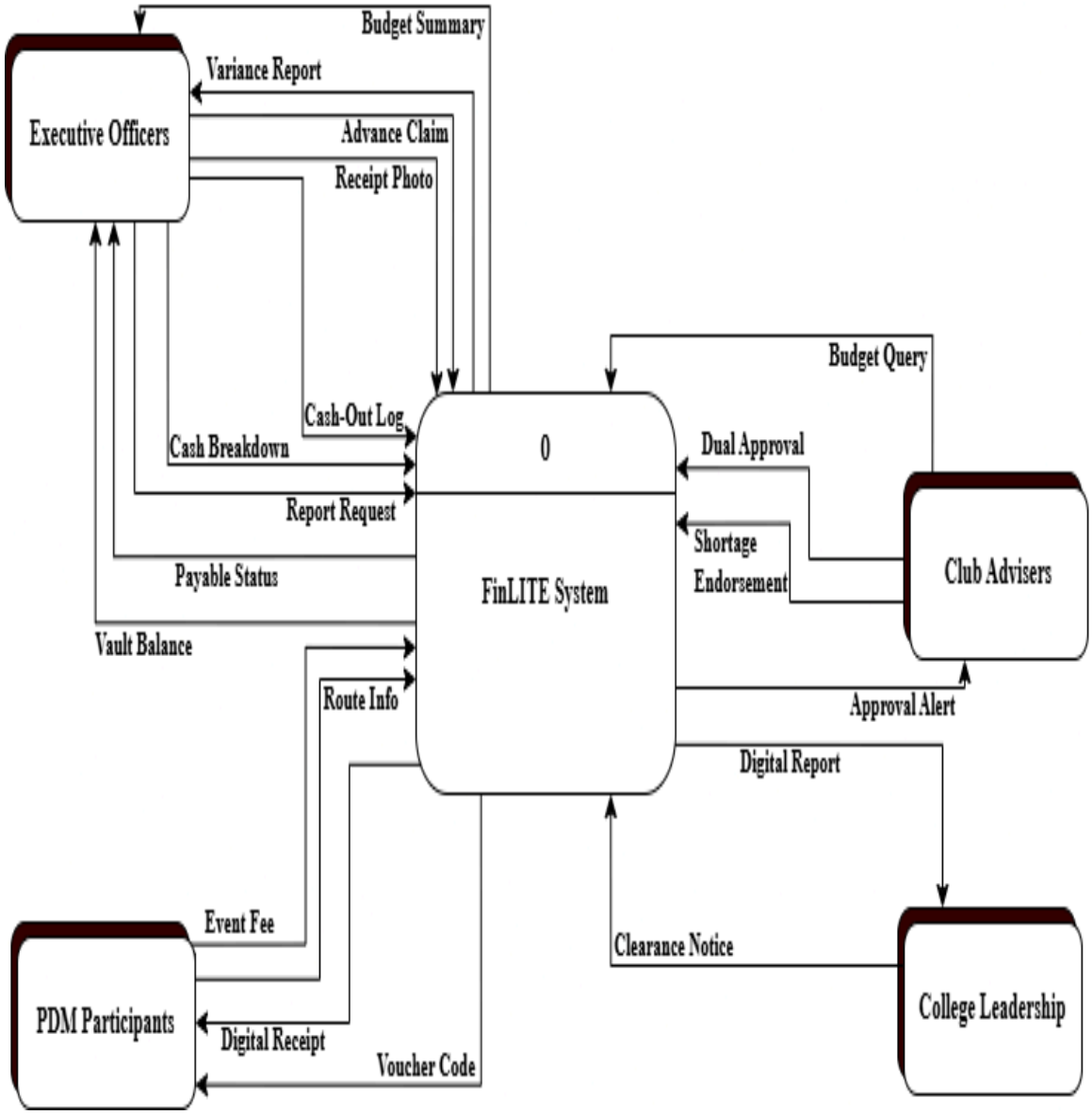


Figure 3: Context Diagram of FinLITE System

Figure 3 illustrates the context diagram of the proposed FinLITE system, showing how the web application centralizes and automates financial management for the student organization. In this setup, executive officers use the system as administrators. Whenever they spend personal money on behalf of the club, they immediately take a photo of the store receipt and file an advance claim online. The platform tracks these out-of-pocket expenses, providing a seven-day countdown timer to ensure officers are reimbursed on time. Officers also log mobile wallet withdrawals along with convenience fees to keep organizational money completely separate from their personal accounts. When students

join club activities or competitions, their registration fees are recorded in the system, and they receive instant digital receipts as electronic proof of payment. In addition, when student representatives run campus errands using tricycles or jeepneys, they enter their travel route directly into the system to generate an electronic petty cash voucher with a tracking code.

During periodic cash reconciliations, officers count the physical bills and coins from the cashbox and type the quantities into the system. FinLITE instantly calculates the total and checks it against the digital ledger, generating a variance report that immediately flags any discrepancies. If an expense exceeds one thousand pesos, the system alerts the faculty club advisers so they can review and submit their digital dual approval. In cases where an unavoidable cash shortage occurs, advisers can also review and endorse the shortage justification online. Advisers and officers can also ask the built-in artificial intelligence chatbot questions to get instant summaries of event budgets and category expenses. Finally, when it is time to submit the end-of-term liquidation documents, the treasurer triggers the report generator with a single click. FinLITE compiles all verified transactions into standardized documents formatted to college standards, which are submitted to the program director and dean for review and final clearance.

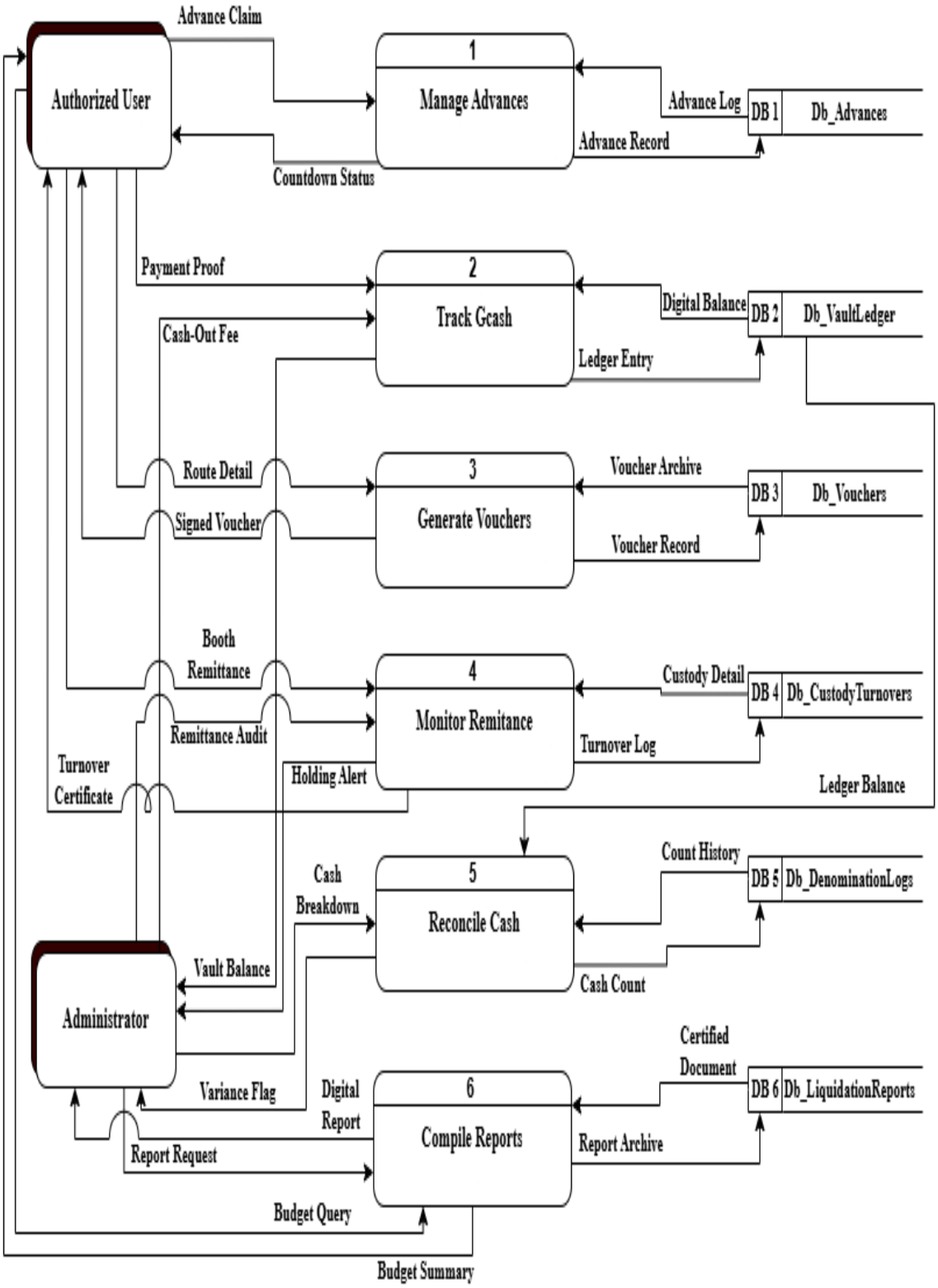


Figure 4: Diagram 0 of FinLITE System

Figure 4 presents the detailed Diagram 0 of the proposed FinLITE system, showing how the central financial process is broken down into six automated modules supported by six dedicated databases. In the first module, when officers spend their own money to buy supplies, they submit their advance claim online along with photos of their receipts. The system saves this record into the advances database and activates a live seven-day countdown timer on their dashboard, ensuring officers are reminded to liquidate their receipts and get refunded promptly. In the second module, digital student payments made through mobile wallets are logged alongside any withdrawal convenience fees. The system saves these entries into the dual vault ledger, clearly separating physical cash on hand from digital funds in transit so that personal money never mixes with club collections. In the third module, whenever student representatives travel for club errands using local tricycles or jeepneys, they enter their travel route presets to generate an electronic petty cash voucher with digital signatures, which is immediately archived into the vouchers database.

In the fourth module, cash collected from event booths is tracked under a strict holding window. If money is held for more than twenty-four to forty-eight hours, the system sends an automatic reminder, and once the funds are turned over, an official turnover reconciliation certificate is generated and stored in the custody database. In the fifth module, during regular cash reconciliations at the adviser desk, officers enter their physical bill and coin counts through an interactive keypad. The system saves these counts into the denomination database, instantly pulls theoretical balances from the ledger, and flags any mathematical discrepancies automatically. In the final module, the treasurer can generate a complete, audit-ready liquidation report with a single click. The platform pulls verified records from all previous databases, formats the document to college standards with proper tables and signatory blocks, and stores the completed report in the liquidation archive. At the same time, student leaders and advisers can type natural questions into the built-in artificial intelligence chatbot to receive instant financial summaries of event budgets and category spending.